Compiled Problems – Intermediate Division

1. **00-01 C4 Prefix/Infix/Postfix**

   Given a quadratic equation in infix form, write it in prefix notation.

   $$AX^2 + BX + C = 0$$

2. **00-01 C4 Prefix/Infix/Postfix**

   For this problem, we define 2 new unary operators: # (absolute value) and % (principal square root). Evaluate the following postfix expression:

   2 6 – # % 3 4 + 2 ↑ % + %

3. **01-02 C4 Prefix/Infix/Postfix**

   Convert the following postfix expression into prefix.

   A A B + C A B / - / *

4. **01-02 C4 Prefix/Infix/Postfix**

   The volume of a sphere is given by the formula $V = \frac{4}{3}\pi R^3$. Translate the formula into prefix.

5. **02-03 C4 Prefix/Infix/Postfix**

   Evaluate the following prefix expression:

   / / * 2 + 4 6 5 - ↑ 3 2 5

6. **02-03 C4 Prefix/Infix/Postfix**

   Convert the following infix expression to postfix:

   $(A + B)^{2/3} + (A / 2 – B) / 3$

7. **03-04 C4 Prefix/Infix/Postfix**

   Convert the following prefix expression to postfix:

   / ↑ + A B 2 * C + ↑ A 2 B

8. 03-04 C4 Prefix/Infix/Postfix

The new binary operation @ is defined as follows: A @ B = $\sqrt{A^2 + B^2}$
Evaluate the following postfix expression. Note that all numbers are single digit.

3 2 ↑ 2 3 * - 2 2 ↑ @ 8 2 – 6 3 / * @

9. 04-05 C4 Prefix/Infix/Postfix

Evaluate the following postfix expression. Note that all numbers are single digit.

8 4 / 6 2 ↑ + 2 + 2 3 ↑ /

10. 04-05 C4 Prefix/Infix/Postfix

Convert the following prefix expression to infix.
Write the answer with the fewest number of parentheses required.

+ / - ↑ A B 2 - C A * A B

11. 05-06 C4 Prefix/Infix/Postfix

Convert the following postfix expression to prefix:

A B + A B - * A 2 - B 2 - - =

12. 05-06 C4 Prefix/Infix/Postfix

Given that A @ B is defined as the Greatest Common Factor (GCF) of A and B, evaluate the following prefix expression using A = 12, B = 40, C = 24, and D = 2.

/ * * @ A B @ A C @ B C * @ A D @ B D

13. 06-07 C4 Prefix/Infix/Postfix

Convert the following prefix expression to postfix:

* / - / A B C + / A B C + B C

14. 06-07 C4 Prefix/Infix/Postfix

Evaluate the following prefix expression given A = 3 and B = 2:

/ - ↑ ↑ A 2 2 ↑ ↑ B 2 2 * - ↑ A 2 ↑ B 2 + ↑ A 2 ↑ B 2

15. 07-08 C4 Prefix/Infix/Postfix

Convert the following postfix expression to prefix:

A B * C A / + A B ↑ - B C / +

16. 07-08 C4 Prefix/Infix/Postfix

Evaluate the following prefix expression. Note the numbers are single digits in the expression.

+ - + 3 / 4 2 / 8 + 3 1 * 6 2

17. 08-09 C4 Prefix/Infix/Postfix

Evaluate the following prefix expression if X = 2:

- + - ↑ X 3 * 3 ↑ X 2 * 3 X 2

18. 08-09 C4 Prefix/Infix/Postfix

Define a @ b = max(a,b). Evaluate the following postfix expression.
Note the numbers are single digits in the expression.

4 2 + 3 / 3 2 8 + * 5 / @ 2 2 ↑ @

19. 09-10 C4 Prefix/Infix/Postfix

Convert the following infix expression to postfix: $\frac{A+B^2}{C} - C*(B+\frac{A}{C})$

20. 09-10 C4 Prefix/Infix/Postfix

Define a # b = greatest common factor of a and b. Evaluate the following prefix expression.

- / + # 12 20 * 6 4 -↑ 2 3 1 / # 150 25 ↑ 5 2

21. 10-11 C4 Prefix/Infix/Postfix

Convert the following infix expression to postfix: $\frac{(A+B)*C}{A*B^2-\frac{B}{C}}$

22. 10-11 C4 Prefix/Infix/Postfix

Define a # b = greatest integer ≤ a/b. Evaluate the following prefix expression.
Note the numbers are single digits.

* + # 9 2 # 8 3 ↑3 2 4

23. 11-12 C2 Prefix/Infix/Postfix

Convert this infix expression into a postfix expression
Remember, all operations must be performed in the same order as the given expression.

4 + 5 * 8 – 6 / 2 – 7

24. 11-12 C2 Prefix/Infix/Postfix

Evaluate this following prefix expression.

+ * 4 ↑– / 6 2 1 2 * / 8 2 + – ↑ 3 2 8 1

25. 12-13 C2 Prefix/Infix/Postfix

Convert this infix expression into a prefix expression.

$A^2 + B * (A - B)^2 - C / A * (B^2 - C)$

26. 12-13 C2 Prefix/Infix/Postfix

Evaluate the following postfix expression:

2 3 9 + * 6 / 8 2 / 6 4 + * 5 / -

27. 13-14 C2 Prefix/Infix/Postfix

Convert this infix expression into a postfix expression: $\frac{A^2(B+C)}{A-C} + \frac{A+B}{C^2}$

28. 13-14 C2 Prefix/Infix/Postfix

Evaluate this following prefix expression. All numbers are single digits.

+ + / * 3 ↑ + 2 8 2 ↑ 5 2 4 / * 6 8 * 4 3

29. 14-15 C2 Prefix/Infix/Postfix

Convert this infix expression into a prefix expression. $A^2 + \frac{2B}{C} + \frac{A}{B+C} - \frac{A}{B}$

30. 14-15 C2 Prefix/Infix/Postfix

Define bdiff , a trinary operator that takes 3 operators a, b c and returns max(abs(a-b), abs(b-c), abs(a-c)) where abs is absolute value. Evaluate this following prefix expression. All numbers are single digits.

bdiff 4 bdiff 3 1 8 6

31. 15-16 C2 Prefix/Infix/Postfix

Convert this infix expression (Heron's formula for the area of a triangle) into a prefix expression.

$(s * (s - a) * (s - b) * (s - c))^{1/2}$

32. 15-16 C2 Prefix/Infix/Postfix

Evaluate the following postfix expression. Note: all numbers are single digits.

2 3 2 ↑ − 4 8 2 / 1 + * 2 / + 2 3 * 6 / +